YUNSEONG MOON (문윤성)

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RESEARCH INTERESTS

- Computer graphics and computational imaging
- Appearance acquisition and modeling, inverse rendering
- Optical imaging systems: polarization, hyperspectral, event, and transient imaging
- 3D imaging and reconstruction

EDUCATION

Pohang University of Science and Technology (POSTECH) *Feb 2024 – Present*
M.S.–Ph.D. Integrated Program, Computer Science and Engineering
Computer Graphics Lab, Advisor: Prof. Seung-Hwan Baek

Pohang University of Science and Technology (POSTECH) *Feb 2019 – Feb 2024*
B.S., Computer Science and Engineering *Magna Cum Laude*

PUBLICATIONS

* Equal contribution

Journal Papers

[J1] Seokjun Choi*, **Yunseong Moon***, Kaizhang Kang, Hoon-Gyu Chung, Jin-Nyeong Kim, Giljoo Nam, Seung-Hwan Baek, *Snapshot Polarimetric Display Inverse Rendering*, **ACM Transactions on Graphics** (Proc. **SIGGRAPH Asia** 2026). [Project Page]

Conference Papers

[C1] Yonghee Oh, Ryota Maeda, **Yunseong Moon**, Seung-Hwan Baek, *Real-world Polarimetric Environment Map Dataset*, **SIGGRAPH Asia** 2026.

[C2] Suhyun Shin, **Yunseong Moon**, Ryota Maeda, David B. Lindell, Kiriakos N. Kutulakos, Seung-Hwan Baek, *Broadband Hyperspectral 3D Imaging under Dispersed Structured Light*, **SIGGRAPH** 2026. [Project Page]

[C3] **Yunseong Moon**, Ryota Maeda, Suhyun Shin, Inseung Hwang, Youngchan Kim, Min H. Kim, Seung-Hwan Baek, *Hyperspectral Polarimetric BRDFs of Real-world Materials*, **SIGGRAPH Asia** 2025. [Project Page]

[C4] Ryota Maeda, **Yunseong Moon**, Seung-Hwan Baek, *Event Ellipsometer: Event-based Mueller-Matrix Video Imaging*, **CVPR** 2025, **Highlight**. [Project Page]

[C5] Yujin Jeon*, Eunsue Choi*, Youngchan Kim, **Yunseong Moon**, Khalid Omer, Felix Heide, Seung-Hwan Baek, *Spectral and Polarization Vision: Spectro-polarimetric Real-world Dataset*, **CVPR** 2024, **Highlight**. [Project Page]

Posters

[P1] Oscar Pueyo-Ciutad, Guillermo Enguita-Lahoz, **Yunseong Moon**, Ryota Maeda, Seung-Hwan Baek, Albert Redo-Sanchez, Diego Gutierrez, *Transient Polarimetry*, **SIGGRAPH** 2026 Posters.

WORK EXPERIENCE

- Part-time Software Engineer, Sketchsoft Inc., Seoul, Republic of Korea *Sep 2022 – Aug 2023*
- Full-time Software Engineer, Sketchsoft Inc., Seoul, Republic of Korea *Jan 2022 – Jul 2022*
- Software Engineer Intern (Full-time), DEMO, Pohang, Republic of Korea *Dec 2020 – Feb 2021*

HONORS AND AWARDS

- BK21 Outstanding Paper Award (1st Prize) *Jan 2026*

TEACHING EXPERIENCE

- Teaching Assistant, Youth AI·Data Science Program – Computer Vision *Nov – Dec 2025*
- Teaching Assistant, CSED520: Computational Imaging, POSTECH *Sep – Dec 2025*
- Teaching Assistant, POSCO AI Program – Computer Vision *Jun 2024*

PROFESSIONAL SERVICE

- Reviewer: ACM Transactions on Graphics (TOG), IEEE Transactions on Visualization and Computer Graphics (TVCG), Asian Conference on Computer Vision (ACCV)

PATENTS

- Method and Apparatus for 3D Modeling, Korea Patent No. 10-2023-0059316 *Registered May 2023*